
📈 Generalized Linear Models 📊

INS-605: Data Analysis II



Content

1 Multiple Linear Regression (MLR)

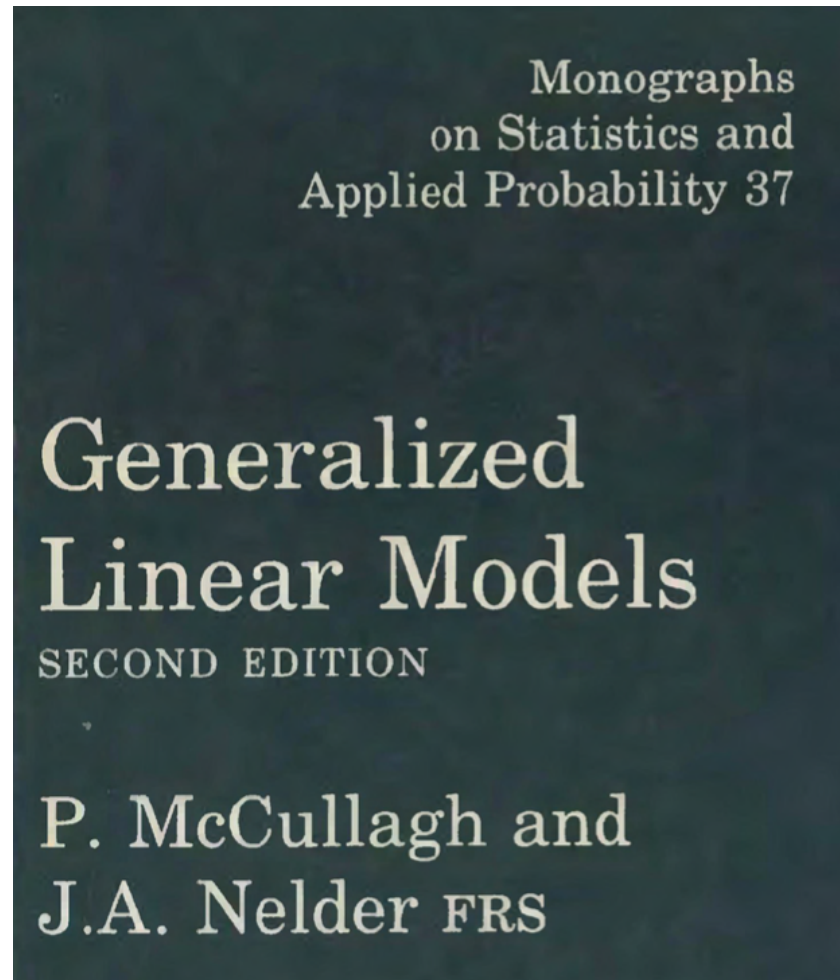
2 Logistic Regression

3 Poisson Regression

4 Statistical Test & CIs

Helpful Resources

🌐 Generalized Linear Models 2nd Edition by P. McCullagh & J.A. Nelder FRS (1983).



What You Will Learn

By the end of this section, you should be able to:

1. Explain the structure of **linear**, **logistic**, and **Poisson** regression.
2. Construct a **linear predictor** $\hat{y}_i = \beta_0 + \beta_1 x_{i1} + \dots + \beta_d x_{id}$.
3. Use a **link function** to connect the *response* to the *linear predictor*.
4. Derive the appropriate *loss function* from the assumed distribution of Y .
5. Interpret coefficients for:
 - Continuous outcomes,
 - Binary outcomes,
 - Count outcomes.
6. Perform statistical tests and confidence interval analysis.
7. Recognize appropriate applications of each model.

1 Multiple Linear Regression (MLR)

1.1 MLR Recap

- Consider **Boston House Price Dataset**:

	CRIM	INDUS	RM	AGE	TAX	PTRATIO	LSTAT	MEDV
0	0.00632	2.31	6.575	65.2	296.0	15.3	4.98	24.0
1	0.02731	7.07	6.421	78.9	242.0	17.8	9.14	21.6
2	0.02729	7.07	7.185	61.1	242.0	17.8	4.03	34.7
3	0.03237	2.18	6.998	45.8	222.0	18.7	2.94	33.4
4	0.06905	2.18	7.147	54.2	222.0	18.7	5.33	36.2

- CRIM**: Per Capita Crime Rate by town
- INDUS**: Proportion of Non-retail Business acres per town
- RM**: Average Number of Rooms per dwelling
- AGE**: Proportion of Owner-occupied units built prior to 1940
- TAX**: Full-value Property-tax rate per 10,000 dollars
- PTRATIO**: Pupil-Teacher Ratio by town
- LSTAT**: Percentage Lower Status of the Population
- MEDV**: Median value of owner-occupied homes in \$1000.

- Notation:

- Individual features: $\underbrace{\mathbf{x}_i}_{[1, \text{features}_i]} \in \mathbb{R}^{d+1}$
- Design matrix of features: $\mathbf{X} \in \mathbb{R}^{n \times (d+1)}$
- Individual target/output: $y_i \in \mathbb{R}$,
- Vector of target: $\mathbf{Y} \in \mathbb{R}^n$.

- Model's coefficients**: $\boldsymbol{\beta} = [\beta_0, \dots, \beta_d] \in \mathbb{R}^{d+1}$

- Predicted point**: $\hat{y}_i = \mathbf{x}_i^T \boldsymbol{\beta} \in \mathbb{R}$
- Predicted Vector**: $\hat{\mathbf{Y}} = \mathbf{X}\boldsymbol{\beta} \in \mathbb{R}^n$.
- Residual/error**: $\mathbf{e}_i = y_i - \hat{y}_i$

Gaussian Linear Model Assumption:

$\mathbf{e}_i \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2)$ for all $i = 1, 2, \dots, n$,
for some fixed std $\sigma > 0$.

- The **residual** should be **centered** around 0 with **constant variance**.

1.1 MLR Recap

- **Ordinary Least Square Goal:** Find the best $\hat{\beta}^*$ by minimizing **Residual Sum of Squares (RSS)**:

$$\hat{\beta}^* = \arg \min_{\beta} \sum_{i=1}^2 (y_i - \mathbf{x}_i^T \beta)^2 = \|Y - X\beta\|^2.$$

- You can check why β is chosen this way!

Min. **RSS** $\Leftrightarrow$ Max. log-likelihood of data.

- **Hint:** $e_i = y_i - \hat{y}_i \sim \mathcal{N}(0, \sigma^2)$ for all i .
 - Step 1: write log-likelihood of e_i 's.
 - Step 2: maximize LLH w.r.t. β .
- Many other loss functions in ML are derived this way!

- Notation:

- Individual features: $\underbrace{\mathbf{x}_i}_{[1, \text{features}_i]} \in \mathbb{R}^{d+1}$
- Design matrix of features: $X \in \mathbb{R}^{n \times (d+1)}$
- Individual target/output: $y_i \in \mathbb{R}$,
- Vector of target: $Y \in \mathbb{R}^n$.

- **Model's coefficients:** $\beta = [\beta_0, \dots, \beta_d] \in \mathbb{R}^{d+1}$

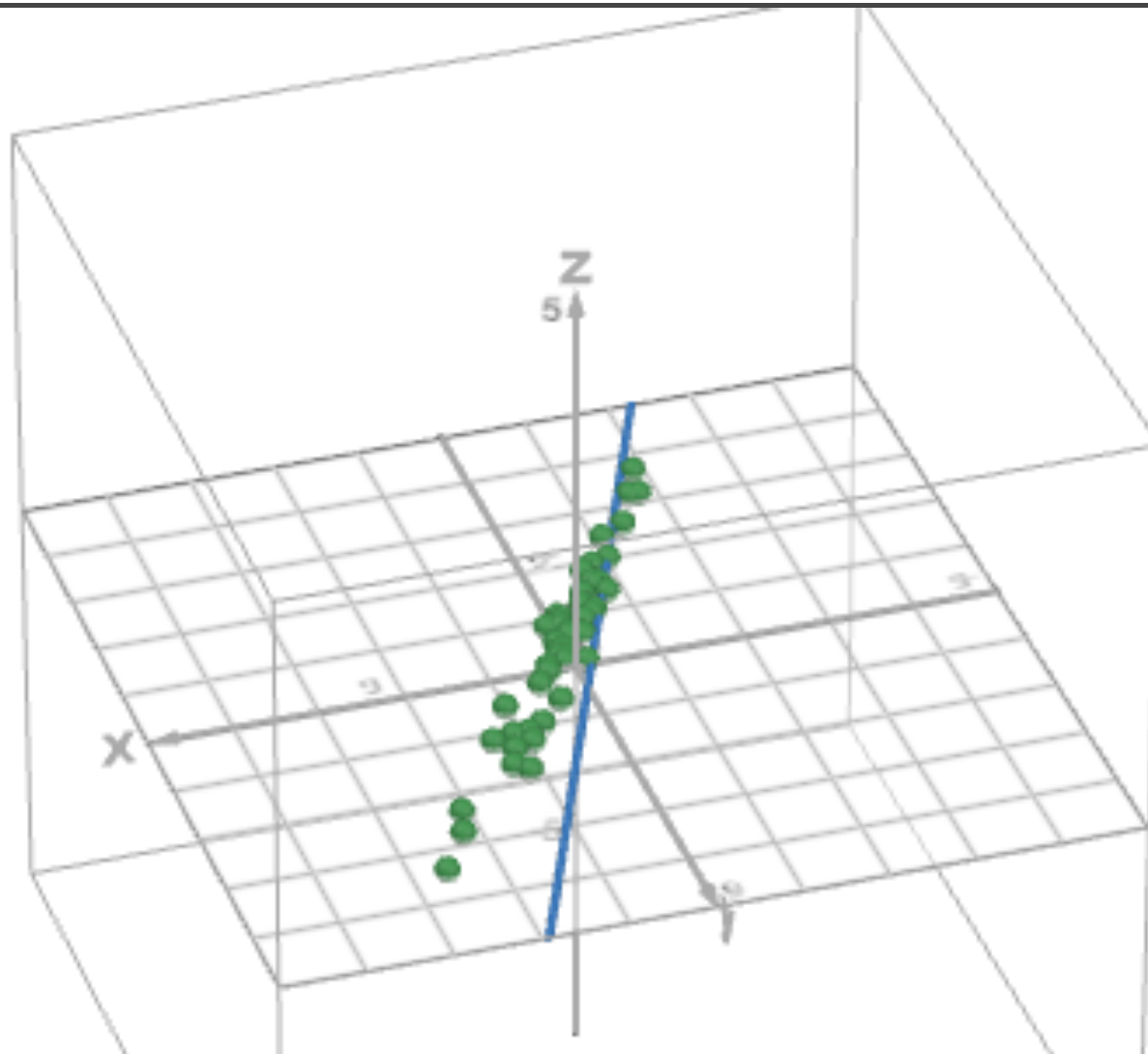
- **Predicted point:** $\hat{y}_i = \mathbf{x}_i^T \beta \in \mathbb{R}$
- **Predicted Vector:** $\hat{Y} = X\beta \in \mathbb{R}^n$.
- **Residual/error:** $e_i = y_i - \hat{y}_i$

Gaussian Linear Model Assumption:

$e_i \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2)$ for all $i = 1, 2, \dots, n$,
for some fixed std $\sigma > 0$.

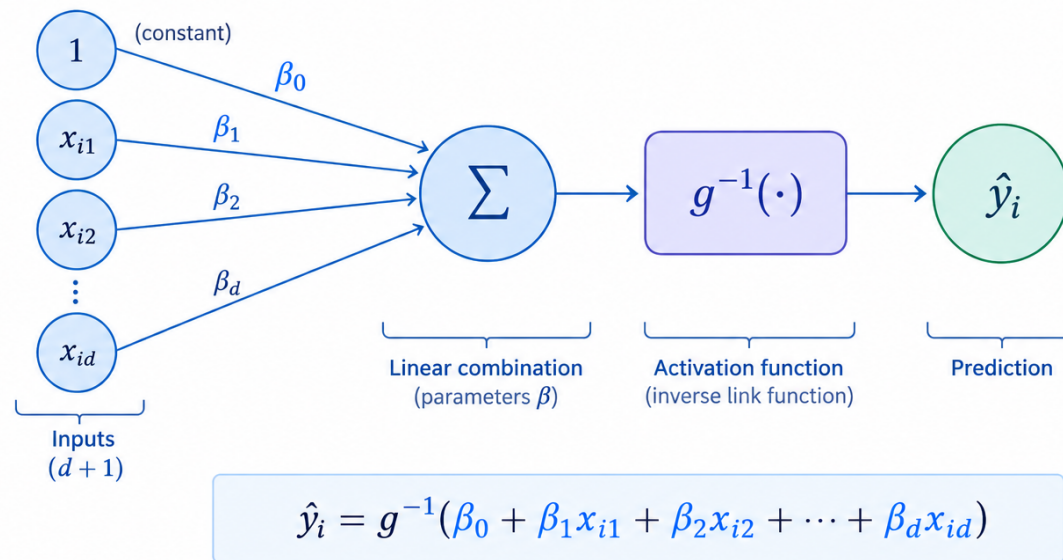
- The **residual** should be **centered** around 0 with **constant variance**.

Visualize **RSS**



1.2 A Different View Of LM

- We can view Linear Model as a simple neural network:



- g^{-1} is called **activation function** in ML, but called **inverse link function** in statistical learning.
- GLM** is all about using suitable g for the corresponding target Y .

Real Target: Multiple Linear Regression

- For GLR (the target $y_i \in \mathbb{R}$) for a fixed input $\mathbf{x}_i \in \mathbb{R}^{d+1}$:

$$\mathbf{e}_i = y_i - \hat{y}_i \sim \mathcal{N}(0, \sigma^2)$$

$$\Leftrightarrow y_i = \hat{y}_i + \mathbf{e}_i \sim \mathcal{N}(\hat{y}_i, \sigma^2)$$

$$\Rightarrow \mathbb{E}[y_i \mid \mathbf{x}_i] = \hat{y}_i = \mathbf{x}_i^T \boldsymbol{\beta} \text{ (also a real number).}$$

Link Function

- A **Link Function** g is used for converting the average value of the target $\mu_i = \mathbb{E}[y_i \mid \mathbf{x}_i]$ to match the **range** of $\hat{y}_i = \mathbf{x}_i^T \boldsymbol{\beta} \in \mathbb{R}$, i.e.,

$$g(\mu_i) = \hat{y}_i = \mathbf{x}_i^T \boldsymbol{\beta} \text{ or } \mu_i = g^{-1}(\mathbf{x}_i^T \boldsymbol{\beta}).$$

- This means that for GLM, $g(\mu) = \mu$ or $g^{-1}(\mu) = \mu$ (identity).

Real Target: Interpretation Of β In MLR

- **MLR** is widely used due to its **interpretability** and **transparency**.

$$y = \beta_0 + \beta_1 x_1 + \cdots + \beta_d x_d + e = \mathbf{x}^T \boldsymbol{\beta} + e.$$

1. None-standardized input (original scale)	2. Standardized input (z-score)	3. One hot encoding input (categorical variable)
$\hat{y} = \beta_0 + \beta_1 x_1 + \cdots + \beta_d x_d$	$\hat{y} = \beta_0 + \beta_1 z_1 + \cdots + \beta_d z_d$ $z_j = \frac{x_j - \mu_j}{\sigma_j}$	$\hat{y} = \beta_0 + \beta_1 D_1 + \beta_2 D_2 + \cdots + \beta_{K-1} D_{K-1}$ $D_k = 1 \text{ if category } k, 0 \text{ otherwise}$
Feature x_j → Coefficient β_j β_j	Feature z_j → Coefficient β_j β_j	Category (dummy) D_k → Coefficient β_k β_k
Interpretation A 1-unit increase in x_j (in its original unit) is associated with a β_j change in $\hat{y}$, holding other features constant.	Interpretation A 1 standard deviation increase in x_j is associated with a β_j change in $\hat{y}$, holding other features constant.	Interpretation β_k is the expected difference in $\hat{y}$ between category k and the reference category (when $D_k = 0$), holding other features constant.

1.3 Application On MDEV Dataset

- Pearson corr:

► Code

	CRIM	INDUS	RM	AGE	TAX	PTRATIO	LSTAT	MEDV
CRIM	1.00	0.41	-0.22	0.35	0.58	0.29	0.46	-0.39
INDUS	0.41	1.00	-0.39	0.64	0.72	0.38	0.60	-0.48
RM	-0.22	-0.39	1.00	-0.24	-0.29	-0.36	-0.61	0.70
AGE	0.35	0.64	-0.24	1.00	0.51	0.26	0.60	-0.38
TAX	0.58	0.72	-0.29	0.51	1.00	0.46	0.54	-0.47
PTRATIO	0.29	0.38	-0.36	0.26	0.46	1.00	0.37	-0.51
LSTAT	0.46	0.60	-0.61	0.60	0.54	0.37	1.00	-0.74
MEDV	-0.39	-0.48	0.70	-0.38	-0.47	-0.51	-0.74	1.00

- Spearman corr:

► Code

	CRIM	INDUS	RM	AGE	TAX	PTRATIO	LSTAT	MEDV
CRIM	1.00	0.74	-0.31	0.70	0.73	0.47	0.63	-0.56
INDUS	0.74	1.00	-0.42	0.68	0.66	0.43	0.64	-0.58
RM	-0.31	-0.42	1.00	-0.28	-0.27	-0.31	-0.64	0.63
AGE	0.70	0.68	-0.28	1.00	0.53	0.36	0.66	-0.55
TAX	0.73	0.66	-0.27	0.53	1.00	0.45	0.53	-0.56
PTRATIO	0.47	0.43	-0.31	0.36	0.45	1.00	0.47	-0.56
LSTAT	0.63	0.64	-0.64	0.66	0.53	0.47	1.00	-0.85
MEDV	-0.56	-0.58	0.63	-0.55	-0.56	-0.56	-0.85	1.00

- **LSTAT** is the strongest correlated feature with **MEDV** for both correlations.
- **CRIM**, **AGE** and **LSTAT** could be engineered as new features (increment in Spearman correlation).
- Feature standarization should be done before fitting the model.

Application On MDEV Dataset

- Performance criteria:

- $\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1} (y_i - \hat{y}_i)^2}.$

- $R^2 = 1 - \frac{\sum_{i=1} (y_i - \hat{y}_i)^2}{\sum_{i=1} (y_i - \bar{y})^2}$: % of variation of Y explained by the model.

► Code

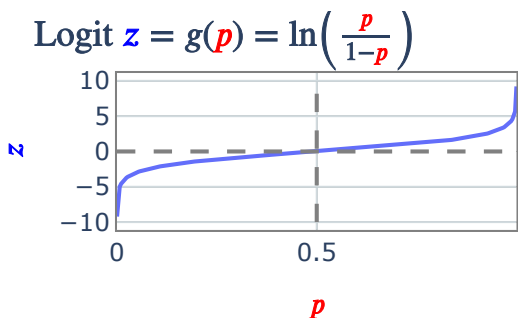
	Intercept	CRIM	INDUS	RM	AGE	TAX	PTRATIO	LSTAT
Coefficient	22.53	-0.49	0.07	3.15	0.56	-0.32	-1.89	-4.12

- R^2 : 0.684.
- $\text{RMSE} / \text{Ave}(Y)$: 0.229.
- $\beta_{\text{CRIM}} = -0.491$ indicates increment in 1-std in **CRIM** leads to around \$500 drop in average median home value.

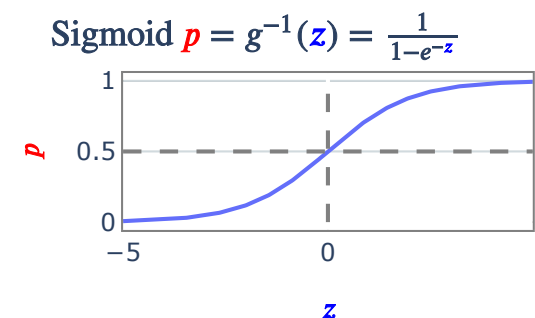
2 Logistic Regression

2.1 Binary Target

- What if the target changes to $y_i \in \{0, 1\}$ such as
 - Sick (1) or healthy (0)
 - Male (1) or female (0)
 - Attack (1) or normal access (0)
- This time $y_i \sim \mathcal{B}(p_i)$ then $\mu_i = \mathbb{E}[y_i \mid \mathbf{x}_i] = p_i \in (0, 1)$.
- What **link function** would map $(0, 1)$ to $\mathbb{R}$?
- **Logit link function**: $g(p) = \ln\left(\frac{p}{1-p}\right)$ is chosen here.

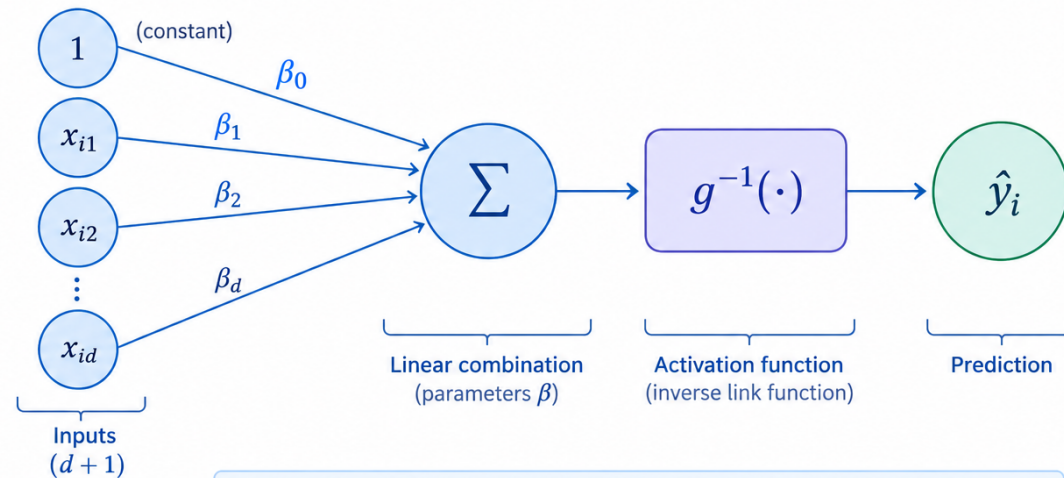


- Its inverse is the well-known **sigmoid** function
$$\sigma(x) = \frac{1}{1+e^{-x}}.$$

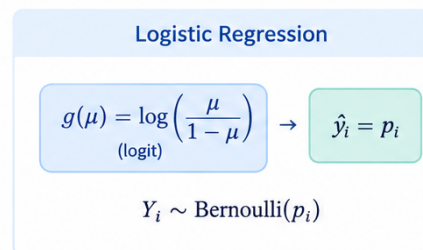


Binary Target: Logistic Regression (Cont.)

- In summary if $y_i \in \{0, 1\}$ then the GLM model is $\hat{y}_i = g^{-1}(\mathbf{x}_i^T \boldsymbol{\beta})$, where $g^{-1}(\mathbf{z}) = \sigma(\mathbf{z}) = \frac{1}{1+e^{-\mathbf{z}}}$.



$$\hat{y}_i = g^{-1}(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_d x_{id})$$



Binary Target: Logistic Regression Loss

- For a binary response, $y_i \sim \mathcal{B}(\hat{y}_i)$ with $\hat{y}_i = \sigma(\mathbf{x}_i^T \boldsymbol{\beta})$.
- Likelihood on data $D = \{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_n, y_n)\}$:

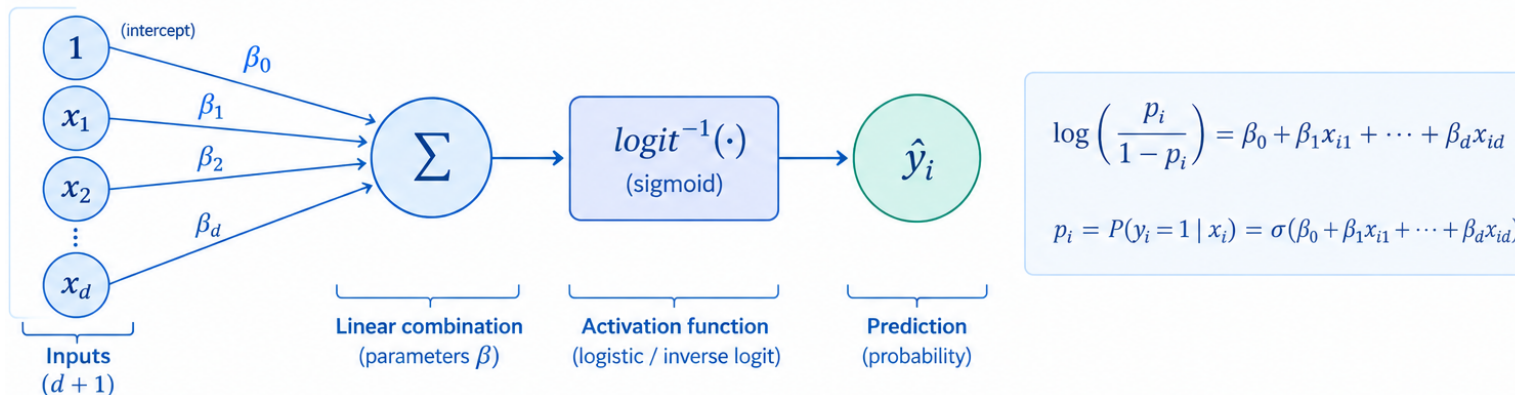
$$\begin{aligned} L(\boldsymbol{\beta}) &= \prod_{i=1}^n \mathbb{P}(Y = y_i) \quad \text{with } P(Y = y_i) = \hat{y}_i^{y_i} (1 - \hat{y}_i)^{1-y_i} \\ &= \prod_{i=1}^n \hat{y}_i^{y_i} (1 - \hat{y}_i)^{1-y_i} \quad (\text{with the convention: } 0^0 = 1) \end{aligned}$$

- Log-likelihood: $\ell(\boldsymbol{\beta}) = \sum_{i=1}^n [y_i \ln(\hat{y}_i) + (1 - y_i) \ln(1 - \hat{y}_i)]$.
- The best parameter $\hat{\boldsymbol{\beta}}^*$ is chosen by maximizing $\ell(\boldsymbol{\beta})$.
- This is equivalent to ML framework: choose the best $\hat{\boldsymbol{\beta}}^*$ by minimizing Binary Cross-Entropy($\boldsymbol{\beta}$) = $-\ell(\boldsymbol{\beta})$.

Binary Target: Coef. β Interpretation

- In binary classification, **odds ratio** = $\frac{\mathbb{P}(y_i=1|X=x_i)}{\mathbb{P}(y_i=0|X=x_i)} = \frac{p_i}{1-p_i}$.

Logistic Regression — Coefficient Interpretation (Log-odds)



Feature	Coefficient (β)	Interpretation (on log-odds)
Intercept (β_0)	Baseline log-odds when all $x = 0$	$\log\left(\frac{p}{1-p}\right)$ for the reference case.
Continuous variable (x_j)	β_j	A 1-unit increase in x_j changes the log-odds by β_j (odds ratio = e^{β_j}).
Binary variable (x_j)	β_j	From 0 to 1, the log-odds changes by β_j (odds ratio = e^{β_j}).
Categorical variable (k levels)	β_k ($k = 1, \dots, K-1$)	Each category k (vs. reference) changes the log-odds by β_k (odds ratio = e^{β_k}).

! Key points

- Coefficients are in log-odds scale (not probability).
- Positive β increases the chance of $y = 1$ (odds ratio > 1).
- Negative β decreases the chance of $y = 1$ (odds ratio < 1).
- The effect size is multiplicative on the odds (e^{β}).



2.2 Example: Exam Passing

- Suppose we model whether a student **passes an exam**:

$$\log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 \text{StudyHours} + \beta_2 \text{Attendance}$$

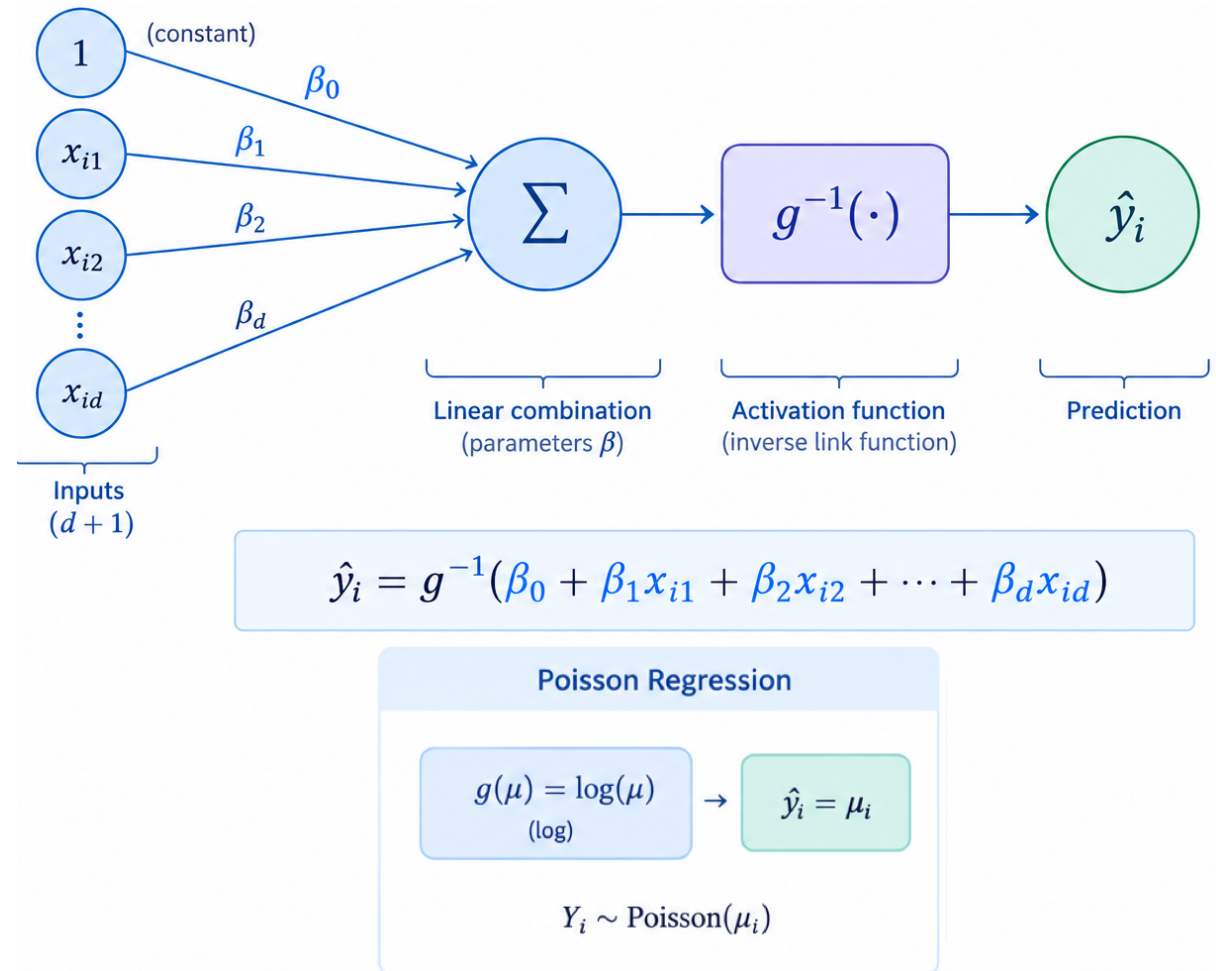
- If $\beta_1 = 0.50$ then
 - **Log-odds**: A 1-hour increase in study time increases the **log-odds of passing** by 0.50, holding **Attendance** constant.
 - **Odds ratio**: Convert the coefficient to an odds ratio: $e^{0.50} \approx 1.65$, meaning that 1-hour increase in study time **multiplies the odds of passing by 1.65**.
 - **Percentage interpretation**: $(1.65 - 1) \times 100\% \approx 65\% \rightarrow$ The odds of passing increase by $\approx 65\%$ for each additional hour of study.
- **Important**: This is a **65% increase in odds**, not a 65%-point increase in **probability**.

3 Poisson Regression

3.1 Count Target

Poisson regression is suitable when the target is integer counts.

- $y_i \sim \mathcal{P}(\lambda_i)$, with $\lambda_i > 0$.
- What **activation (inverse link)** to use?
- Step 1: Compute expectation
 $\lambda_i = \mathbb{E}[y_i \mid X = x_i] > 0$
- Step 2: Which $g : (0, \infty) \rightarrow \mathbb{R}$?
 $g(\lambda) = \ln(\lambda)$.
- Step 3: Activation: $g^{-1}(z) = e^z$.
- **Model:**
$$\hat{y}_i = e^{x_i^T \beta} = e^{\beta_0 + \beta_1 x_{i1} + \dots + \beta_d x_{id}}.$$
- **Dission:** How to estimate β ?



Poisson Coefficient Interpretation

- In Poisson regression: $\ln(\hat{y}_i) = \mathbf{x}_i^T \boldsymbol{\beta} = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_d x_{id}$.

- **Interpretation of coefficients:**

e^{β_j} = Multiplicative change in expected count.

- For example, $\beta_j = 0.10$ implies $e^{0.10} \approx 1.105$. meaning a 1-unit increase in X_j is associated with approximately a **10.5% multiplicative increase in expected count**, holding other variables constant.
- **Predictions:**
 - Raw prediction: $\hat{y}_i = e^{\mathbf{x}_i^T \hat{\boldsymbol{\beta}}^*}$.
 - Rounded prediction: $\text{round}(\hat{y}_i)$.

Model Diagnostics

- **Poisson Deviance** & **Pearson Statistics** are common criteria for assessing the goodness-of-fit of the model:

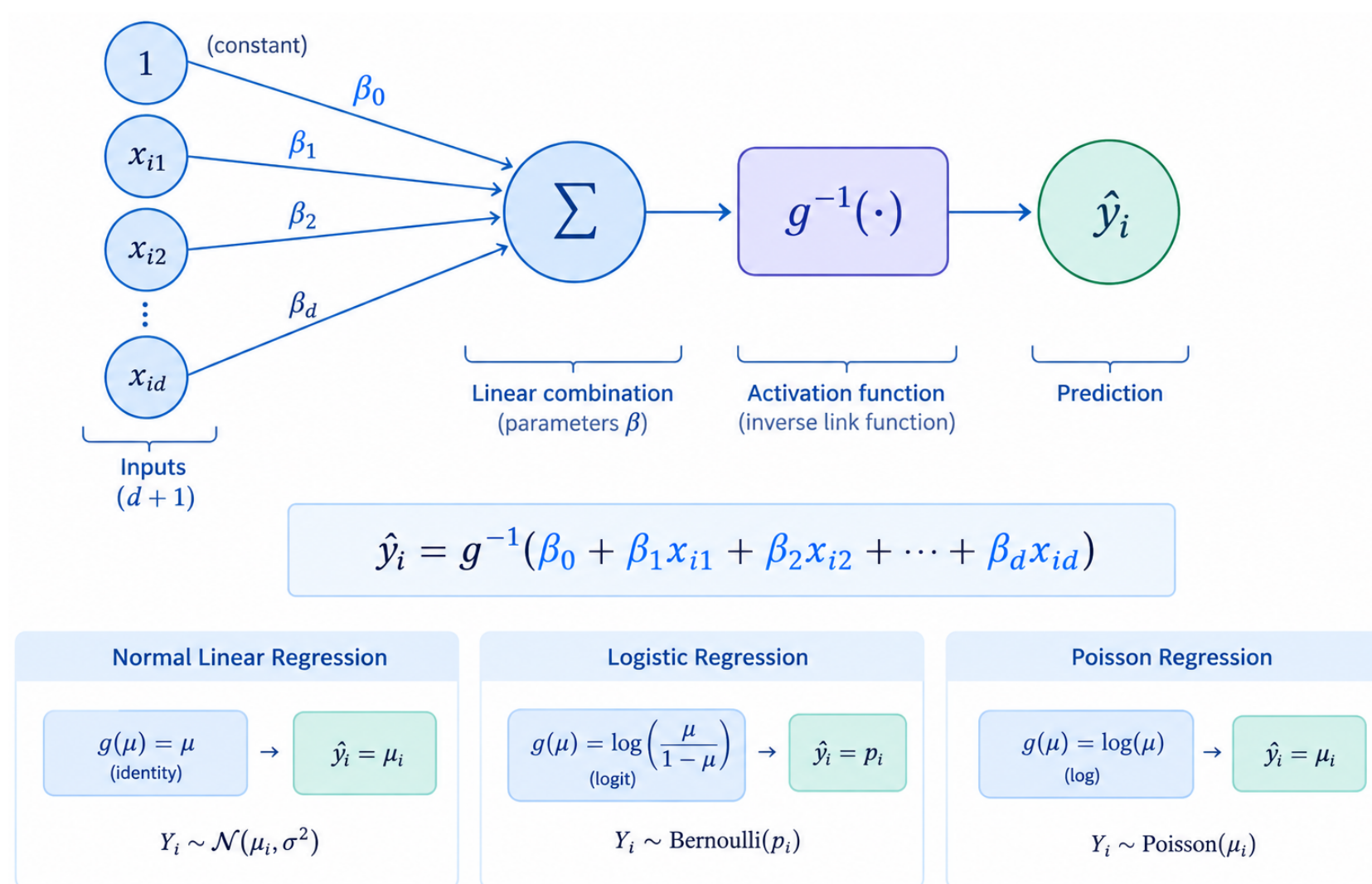
$$D = 2 \sum_{i=1}^n \left[y_i \log \left(\frac{y_i}{\hat{y}_i} \right) - (y_i - \hat{y}_i) \right], \text{ and } X^2 = \sum_{i=1}^n \frac{(y_i - \hat{y}_i)^2}{\hat{y}_i}$$

- D and $X^2 \approx \text{DOF} = n - d - 1$ indicates good fit.
- **Dispersion** for checking dispersion of predictions:

$$\text{Dispersion } \hat{\phi} = \frac{X^2}{n - d - 1}$$

- $\hat{\phi} < 1$: Underdispersed predictions compared to actual data.
- $\hat{\phi} > 1$: Overdispersed predictions compared to actual data.

Summary



Comparison

	Linear	Logistic	Poisson
Target	Continuous	Binary	Count
Distribution	Gaussian	Bernoulli	Poisson
Mean	μ	p	λ
Linear predictor	$X\beta$	$X\beta$	$X\beta$
Link	Identity	Logit	Log
Inverse link	z	$\sigma(z)$	e^z
Loss	MSE	Binary CE	Poisson NLL
Coefficient interpretation	Additive	Odds ratio	Multiplicative

4 Statistical Test & CIs

4.1 Significance Of Coefs & CIs

Statistical Tests and Confidence Intervals

Three Regression Models • Same Workflow, Different Distributions

Multiple Linear Regression (MLR)

Gaussian • Identity Link

$$Y = X\beta + \varepsilon, \quad \varepsilon \sim N(0, \sigma^2)$$

Test Statistic

$$t_j = \frac{\hat{\beta}_j}{SE(\hat{\beta}_j)} \quad df = n - p$$

Confidence Interval

$$\hat{\beta}_j \pm t_{\alpha/2, df} SE(\hat{\beta}_j)$$

Coefficient and SE

$$\begin{aligned} \hat{\beta} &= (X^T X)^{-1} X^T y \\ SE(\hat{\beta}_j) &= \hat{\sigma} \sqrt{[(X^T X)^{-1}]_{jj}} \\ \hat{\sigma}^2 &= \frac{SSE}{n - p} \end{aligned}$$

Logistic Regression

Bernoulli • Logit Link

$$\begin{aligned} P(Y = 1 | X) &= \sigma(X\beta) \\ \sigma(z) &= \frac{1}{1 + e^{-z}} \end{aligned}$$

Test Statistic

$$z_j = \frac{\hat{\beta}_j}{SE(\hat{\beta}_j)} \quad (\text{large-sample})$$

Confidence Interval

$$\begin{aligned} &\hat{\beta}_j \pm z_{\alpha/2} SE(\hat{\beta}_j) \\ (\text{or } &e^{\hat{\beta}_j} \pm z_{\alpha/2} SE(\hat{\beta}_j) \text{ for odds ratio}) \end{aligned}$$

Coefficient and SE

$$\begin{aligned} \hat{\beta} &= (X^T W X)^{-1} X^T (y - \hat{p}) \\ SE(\hat{\beta}_j) &= \sqrt{[(X^T W X)^{-1}]_{jj}} \\ W &= \text{diag}(\hat{p}_i(1 - \hat{p}_i)), \quad \hat{p}_i = \sigma(X_i \hat{\beta}) \end{aligned}$$

Poisson Regression

Poisson • Log Link

$$\begin{aligned} Y &\sim \text{Poisson}(\lambda), \quad \log(\lambda) = X\beta \\ \lambda &= e^{X\beta} \end{aligned}$$

Test Statistic

$$z_j = \frac{\hat{\beta}_j}{SE(\hat{\beta}_j)} \quad (\text{large-sample})$$

Confidence Interval

$$\begin{aligned} &\hat{\beta}_j \pm z_{\alpha/2} SE(\hat{\beta}_j) \\ (\text{or } &e^{\hat{\beta}_j} \pm z_{\alpha/2} SE(\hat{\beta}_j) \text{ for rate ratio}) \end{aligned}$$

Coefficient and SE

$$\begin{aligned} \hat{\beta} &= (X^T W X)^{-1} X^T (y - \hat{\lambda}) \\ SE(\hat{\beta}_j) &= \sqrt{[(X^T W X)^{-1}]_{jj}} \\ W &= \text{diag}(\hat{\lambda}_i), \quad \hat{\lambda}_i = e^{X_i \hat{\beta}} \end{aligned}$$

Diagnostics: What Else To Check?

- Diagnostics is also required when we move from LR to GLMs.
- **1. Is the model appropriate?**
 - Is the response distribution reasonable?
- **2. Are important predictors missing?**
 - Use DA to detect useful variables,
 - Nonlinear effects (feature engineering),
 - Interactions (mixture of variables).
- **3. Are there problematic observations?**
 - Outliers,
 - High-leverage observations,
 - Influential observations.

4.4 GLM Summary

- The entire **GLM** framework can be summarized as

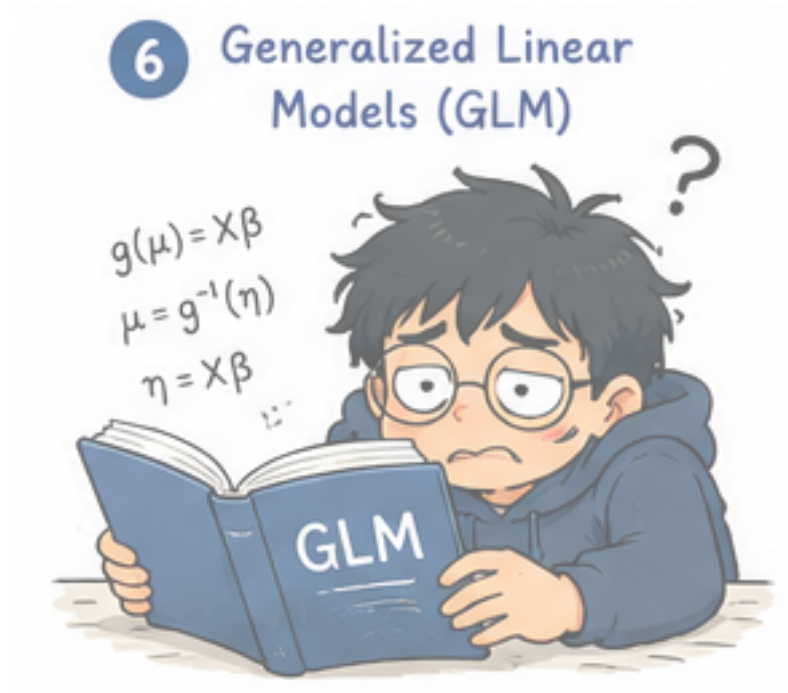
$$X \longrightarrow X\beta \longrightarrow g^{-1}(X\beta) \longrightarrow \mu = E[Y | X].$$

- Main types:

Continuous $\rightarrow$ Linear Regression
Binary $\rightarrow$ Logistic Regression
Count $\rightarrow$ Poisson Regression

- The parameter β is estimated using MLE that leads to different types of loss function in ML.

🥳 Yeahhhh 🍷 !!!



Let's take a break!

Appendix 1

- The Poisson model $y_i \sim \mathcal{P}(\widehat{y}_i)$ with $\widehat{y}_i = e^{\mathbf{x}_i^T \boldsymbol{\beta}}$. The Poisson PMF:

$$P(Y = y_i) = \frac{e^{-\widehat{y}_i} \widehat{y}_i^{y_i}}{y_i!}.$$

- Log-likelihood: $\ell(\boldsymbol{\beta}) = \sum_i \left[-\log(y_i!) - \widehat{y}_i + y_i \log(\widehat{y}_i) \right]$.
- Using $\log(\widehat{y}_i) = \mathbf{x}_i^T \boldsymbol{\beta}$ one has the negative log-likelihood, ignoring

constants, becomes

$$\text{Poisson loss}(\boldsymbol{\beta}) = \sum_{i=1}^n \left[e^{\mathbf{x}_i^T \boldsymbol{\beta}} - y_i \mathbf{x}_i^T \boldsymbol{\beta} \right].$$

- One can learn the best parameters $\boldsymbol{\beta}$ by minimizing Poisson loss.